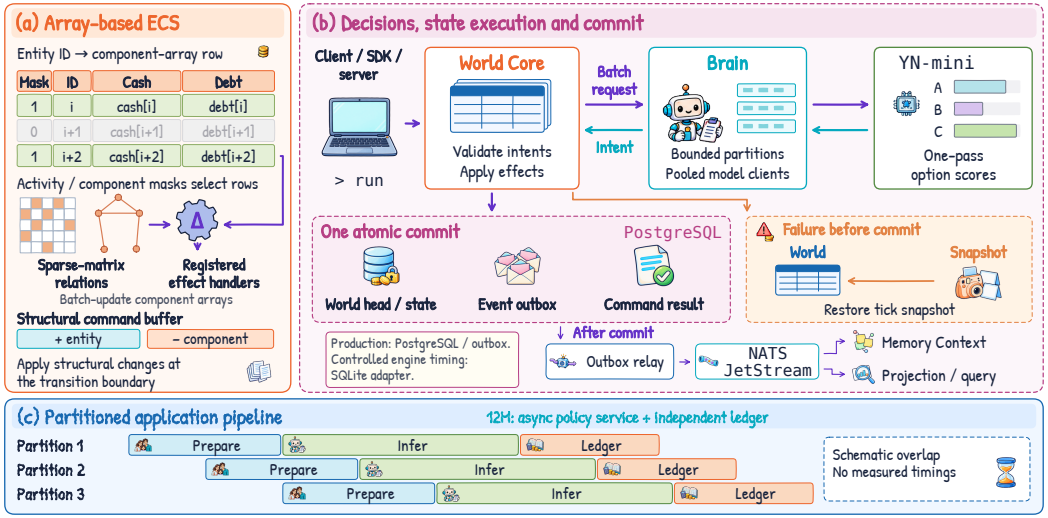

(a) Array-based ECS
Entity ID → component-array row
Mask
ID
Cash
Debt
1
i
cash[i]
debt[i]
0
i+1
cash[i+1]
debt[i+1]
1
i+2
cash[i+2]
debt[i+2]
Activity / component masks select rows
Sparse-matrix relations
Registered effect handlers
Batch-update component arrays
Structural command buffer
+ entity
- component
Apply structural changes at the transition boundary
(b) Decisions, state execution and commit
Client / SDK / server
> run
World Core
Validate intents
Apply effects
Batch request
Intent
Brain
Bounded partitions
Pooled model clients
YN-mini
A
B
C
One-pass option scores
One atomic commit
PostgreSQL
World head / state
Event outbox
Command result
Failure before commit
World
Snapshot
Restore tick snapshot
Production: PostgreSQL / outbox.
Controlled engine timing:
SQLite adapter.
After commit
Outbox relay
NATS JetStream
Memory Context
Projection / query
(c) Partitioned application pipeline
12M: async policy service + independent ledger
Partition 1
Partition 2
Partition 3
Prepare
Infer
Ledger
Schematic overlap
No measured timings